

# MOYAHABO THABANG RANTOKA

## COMPUTER SYSTEMS ENGINEERING | Full Stack Developer

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Portfolio: <https://thabangrantoka.github.io/thabang-portfolio/>

### PROFESSIONAL PROFILE

Technically astute Computer Systems Engineering graduate with hands-on experience in data architecture, systems integration, and business process automation. Through my Full Stack Developer Internship at MMS Africa and academic projects, I have developed a strong ability to translate complex business requirements into scalable technical solutions (e.g., role-based operations platforms, API integrations, and Power BI dashboards). I possess a sharp analytical mindset, a proven track record of high performance in Agile environments, and a passion for leveraging data and technology to solve real-world challenges in the financial and insurance sectors.

### CORE TECHNICAL SKILLS

**Frontend Development:** HTML5, CSS3, JavaScript, React, Angular, Responsive Design, Bootstrap

**Backend Development:** Python, PHP, Node.js (Learning), Flask, RESTful API Development, Server-side Development

**Database & Data Architecture:** SQL, MySQL/MariaDB, PostgreSQL (Learning), MongoDB (Learning), Database Design & Optimization

**Systems Integration:** Google APIs, Facebook Graph API, Salesforce Integration, Third-party API Integration

**Tools & Frameworks:** Git, GitHub, Power BI, XAMPP, Docker (Learning), Microsoft Azure

**Additional Skills:** Embedded Systems, Raspberry Pi, Arduino, Robotics, PLC Fundamentals

**Professional Skills:** Problem Solving, System Integration, Technical Documentation, Team Collaboration, Software Testing & Debugging

### EDUCATION

#### Tshwane University of Technology — Pretoria, Gauteng

Diploma in Computer Systems Engineering | January 2022 – June 2026

- Relevant modules: Network Systems, Operating Systems, Software Engineering, Programming, Databases, Embedded Systems, Artificial Intelligence, Mobile Robotics, Electronics, Digital Process Control, Programmable Logic Controllers and Project Design.

#### George Tladi Technical High School — Mohodi, Limpopo

National Senior Certificate / Grade 12 | 2016 – 2020

- Relevant subjects: Mathematics, Physical Sciences, and Engineering Graphics and Design.

### PROFESSIONAL EXPERIENCE

#### Full Stack Developer Intern MMS Africa – RANDBURG, Gauteng 2025 – Present (Completing SEPTEMBER 2026)

MMS Africa is a leading broadcast software development company specializing in media management solutions for broadcasters across Africa.

##### Key Responsibilities & Achievements:

- **Backend & API Development:** Developed and maintained robust backend services and RESTful APIs using JavaScript and PHP, ensuring seamless data transmission and system reliability.
- **Frontend Development:** Designed and implemented responsive, cross-browser-compatible user interfaces using Angular, focusing on high usability and intuitive user experiences.
- **Data Architecture & Integration:** Created optimized SQL database schemas, integrated third-party APIs (Google, Facebook, Salesforce), and built interactive Power BI dashboards for data analysis.

- **Systems Integration:** Successfully connected legacy systems with modern applications by developing custom middleware and implementing secure authentication protocols.
- **Agile Collaboration & Quality Assurance:** Actively participated in Agile sprints, peer code reviews, and project tracking. Authored comprehensive technical documentation and conducted rigorous testing to ensure solutions were maintainable, re-usable, and met business requirements.

## PROJECT EXPERIENCE

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### OpsFlow — Business Operations Management System | *PHP 8, MySQL, Git, CSS, XAMPP*

- Developed a comprehensive, role-based operations platform featuring employee/department CRUD, project tracking, attendance, and leave approvals.
- Implemented secure bcrypt password hashing and a strict Admin/Manager/Employee permission matrix to ensure data security and access control.
- Built live KPI dashboards and automated CSV reporting, providing actionable data insights for management decision-making.
- Maintained detailed admin activity logs to ensure system traceability and accountability, aligning with enterprise security best practices.

### BrainWave Work AI | *TypeScript, React, Tailwind CSS, TanStack Start*

- Engineered a modern AI productivity workspace featuring an email generator, meeting-notes summarizer, task planner, and integrated AI chat.
- Applied a "human-in-the-loop" design philosophy, ensuring AI generates drafts while maintaining professional oversight and control.
- Developed a responsive, highly performant frontend using React and Tailwind CSS, backed by an analytics dashboard for usage tracking.

### AI Home Automation & Robotics Systems | *Python, Flask, JavaScript, Unity ML-Agents, Arduino*

- Built a full-stack Flask web interface to remotely control Raspberry Pi GPIO relays, integrating hardware and software seamlessly.
- Developed and trained Unity ML-Agents for autonomous mobile robot navigation (obstacle avoidance and wall-following), applying reinforcement learning concepts to real-world hardware.

### Municipal Billing System | *C++, Object-Oriented Programming (OOP), File Handling*

- Designed a robust desktop application to automate customer accounts, tariff management, and billing calculations.
- Applied strict Object-Oriented Programming (OOP) principles to create a modular, readable, and maintainable billing engine.
- Conducted rigorous testing to ensure accurate tariff calculations and reliable file-based data storage.

## PROFESSIONAL STRENGTHS

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- Technical problem solving and troubleshooting
- Analytical thinking and attention to detail
- Fast learning and adaptability
- Communication and teamwork
- Time management and critical thinking
- Ability to work across hardware and software environments

## CERTIFICATIONS & LICENCE

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- Introduction to Python — Cisco Networking Academy, 2023
- Code 10 Driver's Licence
- **Google AI Essential**
- **FNB Academy CERTIFICATE**

LANGUAGES

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English | Sepedi | isiZulu